

ALGORITHM – STUDENT MANAGEMENT SYSTEM (WEEK 1)

Purpose: To manage student records by using basic operations like Add, View, Update and Delete using ArrayList and Encapsulation.

1 Start

- Start the program.

2 Create ArrayList

- Create an ArrayList to store Student objects.

3 Display Menu

- Display the menu with the following options:

- 1. Add Student
- 2. View Students
- 3. Update Student
- 4. Delete Student
- 5. Exit

4 Read Choice

- Read the user's choice.

5 Add Student

- If choice is 1:
 - Read student details (ID, Name, Department).
 - Create a Student object.
 - Add the object to the ArrayList.

6 View Students

- If choice is 2:
 - If ArrayList is empty, display “No students found”.
 - Otherwise, display all student details.

7 Update Student

- If choice is 3:
 - Read the ID of the student to update.
 - Search the student in the ArrayList.
 - If found, read new details and update the record using setter methods.
 - Otherwise, display “Student not found”.

8 Delete Student

- If choice is 4:
 - Read the ID of the student to delete.
 - Search the student in the ArrayList.
 - If found, remove the student from the ArrayList.
 - Otherwise, display “Student not found”.

9 Exit

- If choice is 5, terminate the program.

10 Repeat

- Repeat steps 3 to 9 until the user chooses to exit.

11 End

- Stop the program.

